

CSP 571

DATA PREPARATION AND ANALYSIS

PROJECT GROUP AND TOPIC

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PROJECT TOPIC:

DIAGNOSIS OF POLYCYSTIC OVARY SYNDROME

Project Report: Machine LearningBased Diagnosis of Polycystic Ovary Syndrome (PCOS)

1. Abstract

Polycystic Ovary Syndrome (PCOS) is a complex hormonal disorder that affects approximately 8% of women during their reproductive years. It presents a wide range of symptoms, including irregular menstrual cycles, acne, weight gain, and infertility. These symptoms often overlap with other medical conditions, making PCOS challenging to diagnose accurately and promptly. Traditional diagnostic approaches typically involve a process of elimination ruling out other possible disorderswhich can be timeconsuming, subjective, and prone to inconsistencies. To address this, the study aimed to leverage machine learning (ML) techniques for early and reliable diagnosis of PCOS. The primary objectives were to identify the most critical clinical and physical indicators of the condition, build models that are both highly accurate and easy for healthcare professionals to interpret, and apply feature selection methods to reduce the number of input variablesthereby increasing efficiency and transparency. Through univariate feature selection, the study identified the FSH/LH ratio and Body Mass Index (BMI) as the most predictive indicators of PCOS. Among the models developed, a hybrid approach combining Random Forest and Logistic Regression (RFLR) delivered the best performance, achieving an accuracy of 91% and a recall of 90%, which is especially important in minimizing missed diagnoses. To enhance trust and usability in clinical settings, explainability tools such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable ModelAgnostic Explanations) were employed, allowing clinicians to understand the reasoning behind the model's predictions.

2. Overview

2.1 Problem Statement

Polycystic Ovary Syndrome (PCOS) is a complex and heterogeneous hormonal disorder that affects an estimated 6% to 8% of women during their reproductive years. What makes PCOS particularly challenging to diagnose and manage is its **multifaceted nature**meaning it presents in many different ways across different individuals. For example, some women may experience **irregular or missed menstrual periods**, while others may suffer from **excessive hair growth (hirsutism)**, **hair thinning or loss on the scalp**, **acne**, or **skin darkening** in specific areas like the neck or underarms. Additionally, symptoms like **weight gain** and **difficulty conceiving** may also appear, but not all women exhibit all symptoms.

This **variation in symptom expression** makes it difficult for healthcare professionals to identify a clear diagnostic pattern. Often, PCOS is diagnosed by **excluding other potential causes** of these symptoms, such as thyroid disorders or hormonal imbalances caused by other conditions. This method, known as an **exclusionary diagnosis**, is inherently **subjective**, relying heavily on clinical judgment and interpretation rather than objective, quantifiable indicators. As a result, diagnosis may be **delayed or incorrect**, causing frustration for patients and potentially worsening their condition.

Moreover, delayed diagnosis can have **serious longterm consequences**. Untreated PCOS is associated with an increased risk of **infertility**, **type 2 diabetes**, **metabolic syndrome**, **obesity**, **sleep apnea**, and even **cardiovascular disease**. Therefore, **early and accurate detection** is crucial to prevent or manage

these complications effectively. However, the current diagnostic approach is often **timeconsuming**, requiring multiple tests over several visits, and **costly**, particularly in healthcare systems where diagnostic imaging and hormonal assays are expensive or not easily accessible.

This highlights the **urgent need for more efficient, objective, and scalable diagnostic tools** such as machine learningbased modelsthat can assist clinicians in identifying PCOS earlier and with greater accuracy, ultimately improving patient outcomes.

2.2 Literature Survey

Recent research in the field of medical diagnostics has shown promising results in applying **machine learning (ML) techniques** to assist in the diagnosis of **Polycystic Ovary Syndrome (PCOS)**. These studies utilize datasets that include both **clinical parameters** (such as hormone levels like FSH, LH, insulin, and TSH) and **physical attributes** (such as BMI, weight, and skin or hair conditions). By analyzing these diverse variables, ML models are able to uncover complex patterns and relationships that may not be easily noticeable through traditional diagnostic methods.

Among the various ML approaches, **ensemble methods** which combine the predictions of multiple simpler models to improve accuracy have shown particularly strong performance. Two such ensemble techniques are **Gradient Boosting** and **Random Forest**. These algorithms are capable of handling large, highdimensional datasets and are resistant to overfitting. In the context of PCOS diagnosis, they have been reported to achieve **classification accuracies exceeding 85%**, which is considered excellent for medical diagnosis tasks.

A crucial part of building effective ML models for PCOS diagnosis is **feature selection**the process of identifying the most relevant variables (features) from the dataset that contribute significantly to predicting the outcome (i.e., whether a patient has PCOS or not). This step improves model efficiency and helps avoid noise from irrelevant or redundant features. In almost every related study, the **FSH/LH ratio** (the ratio between FollicleStimulating Hormone and Luteinizing Hormone) has consistently been recognized as one of the most **critical indicators** for PCOS. This aligns with medical literature, as a low FSH/LH ratio is a common hormonal imbalance in women with PCOS and reflects disruptions in ovulation. These findings collectively support the integration of machine learning in clinical workflows to support faster and more accurate PCOS diagnosis.

2.3 Proposed Methodology:

The methodology involves:

1. **Data Preprocessing** : Handling missing values, encoding categorical variables, and normalizing numerical features.
2. **Feature Selection** : Using univariate feature selection to identify the top 10 most predictive features.
3. **Model Training** : Applying multiple classifiers (Logistic Regression, Random Forest, Gradient Boosting, LightGBM, KNN, Voting Classifier, RFLR).
4. **Evaluation Metrics** : Assessing models using accuracy, precision, recall, F1score, and crossvalidation.

5. Explainability : Employing SHAP and LIME for interpretable predictions.

3. Data Processing

Data processing is a critical phase in any machine learning project as it ensures that the dataset used to train models is clean, structured, relevant , and conducive to accurate predictions. Below is a comprehensive breakdown of our data processing pipeline and its key components.

3.1 Pipeline Details

We followed a multistage preprocessing workflow tailored to the specific characteristics of the PCOS dataset:

3.1.1 Data Merging

- We merged two related datasets using patient file number as the common identifier.
- This step was necessary because the original data was split into separate files covering different types of clinical parameters.
- Postmerging, we removed redundant or overlapping columns to ensure data consistency and reduce noise.

3.1.2 Data Cleaning

- **Handling Missing Values :**
A few features like TSH , Prolactin , and Insulin levels had missing entries.
To preserve the integrity of the dataset, we filled these missing values using median imputation , which is robust against outliers compared to meanbased methods.
- **Encoding Categorical Variables :**
Several variables such as "Hair loss" , "Skin darkening" , and "Pcos (Yes/No)" were encoded using label encoding , converting stringbased categories into numerical values (e.g., "Yes" = 1, "No" = 0). This made the dataset compatible with most machine learning algorithms that require numerical inputs.
- **Normalization :**
Numerical features such as BMI , FSH , LH , and Age were normalized using MinMax scaling , transforming them to a range between 0 and 1. Normalization ensured that all features contributed equally during model training and prevented certain attributes from dominating others due to scale differences.
- **Train/Test Splitting :**
The final dataset was divided into:
 - 70% training set
 - 30% testing set

- This allowed us to evaluate how well the model generalized to unseen data and avoid overfitting.

➤ **CrossValidation :**

For more rigorous evaluation, we applied 40fold crossvalidation , especially when tuning hyperparameters and comparing model performance. Crossvalidation helped us estimate model accuracy across multiple random splits of the dataset, ensuring stability and reducing variance in performance metrics.

3.2. Data Issues Identified

During the initial exploration and preprocessing stages, we identified several challenges within the dataset that could affect the performance and reliability of the trained models:

3.2.1. Missing Values in HormoneRelated Features

- Certain hormonerelated columns like TSH , Prolactin , and Insulin levels had missing values.
- These missing values could introduce bias if ignored or handled improperly.
- We chose median imputation as it is less affected by extreme values than mean imputation, especially in small medical datasets.

3.2.2. StringBased Categorical Variables

- Many symptoms and physical indicators were represented as strings (e.g., "Yes"/"No").
- Such categorical data must be converted to numeric format before being input into machine learning pipelines.
- We performed label encoding to transform these binary and nominal features into interpretable numbers while maintaining the structure of the dataset.

3.2.3. High Dimensionality (43 Features)

- The raw dataset contained 43 features , making it complex and computationally heavy for training.
- Not all features are equally important; some may even be redundant or noisy.
- We addressed this issue through univariate feature selection , identifying the top 10 most predictive features for use in model training.

3.2.4. Class Imbalance

- Among the 541 patients, only 177 were diagnosed with PCOS , while 364 were not , resulting in a moderate imbalance.
- While not severe, this imbalance could still skew models toward predicting the majority class more often.

- We evaluated model performance using Recall and F1Score to compensate for this imbalance.

3.3. Assumptions and Adjustments Made

To manage the limitations and ensure model effectiveness, we made several assumptions and adjustments:

3.3.1. Missing Data Assumption

- We assumed that missing data occurred randomly and did not reflect a systematic pattern tied to disease progression.
- Based on this assumption, median imputation was deemed appropriate to maintain statistical balance without introducing overfitting risks.

3.3.2. Dataset Representativeness

- We assumed that the Kaggle dataset was sufficiently representative of typical PCOS cases among women of reproductive age.
- Although the demographic scope was limited, the dataset provided a solid foundation for building a preliminary diagnostic tool.
- Future work will include expanding the dataset with diverse international sources.

3.3.3. No External Validation

- Currently, the model has been validated internally using holdout testing and crossvalidation.
- External validation applying the model to independent or realworld data was not included in this phase but is planned for future deployment testing in clinical settings.

3.4. Impact of Preprocessing on Model Development

The preprocessing steps significantly improved the quality of the dataset and enhanced model performance:

STAGE	IMPACT	
Merging Datasets	Ensured comprehensive coverage of clinical and physical features	
Missing Value Handling	Improved data completeness and reduced bias during training	
Feature Encoding	Enabled compatibility with ML algorithms and proper interpretation	
Normalization	Prevented feature dominance and improved convergence speed	
Feature Selection	Reduced dimensionality and computational load without sacrificing accuracy	
Train/Test Split & CV	Enhanced model generalizability and reliability	

These steps laid the groundwork for effective model training and evaluation by ensuring that the dataset was clean, consistent, and optimized for predictive modeling.

4. Data Analysis

Summary Statistics

FEATURE	MEAN (PCOS+)	MEAN (PCOS-)
BMI	28.2	25.1
FSH	6.2	6.5
LH	10.2	6.8
FSH/LH	0.6	1.0

Feature Extraction

In this phase, the goal was to identify which features (variables) in the dataset are most relevant to diagnosing PCOS. The original dataset contained 43 features ranging from hormonal indicators to physical characteristics. However, not all of them contribute equally to prediction, and using too many can cause overfitting and increase computational complexity. To address this, univariate feature selection was applied. This method evaluates each feature individually against the target label ("PCOS: Yes/No") to determine its predictive power.

The top 10 features selected were:

- FSH/LH Ratio – A key hormonal marker; a lower ratio is often linked to PCOS.
- BMI – Body Mass Index; higher BMI values are commonly observed in PCOS patients.
- TSH – Thyroid Stimulating Hormone; imbalances may be linked with PCOS symptoms.
- Prolactin – Elevated levels can mimic PCOS-like symptoms.
- Testosterone – Higher levels indicate hyperandrogenism, a hallmark of PCOS.
- Weight and Height – Used in BMI calculation; also serve as standalone indicators.
- Insulin Levels – Many PCOS patients show signs of insulin resistance.
- Skin Darkening – A visible symptom often associated with insulin resistance.
- Hair Loss – A physical manifestation of hormonal imbalance.

These features were used for training the ML models. By focusing on the most predictive variables, the study ensured that the models were both efficient and accurate.

5. Model Training

Feature Engineering

Feature engineering is the process of preparing and selecting the most relevant variables (features) that will be used as inputs to machine learning models. In this study, feature engineering was done with two primary goals in mind: to maximize diagnostic accuracy and to reduce computational complexity.

➤ **Selected Top 10 Features**

From an initial pool of 43 features in the dataset, a univariate feature selection technique was applied to identify the top 10 most informative features. These included clinical measurements such as the FSH/LH ratio, BMI, testosterone levels, and physical indicators like skin darkening and hair loss. Selecting only the most important features helped the model:

- ❖ Run faster by reducing dimensionality.
- ❖ Avoid learning from irrelevant or noisy data.
- ❖ Prevent overfitting by simplifying the model.

➤ **Handled Multicollinearity**

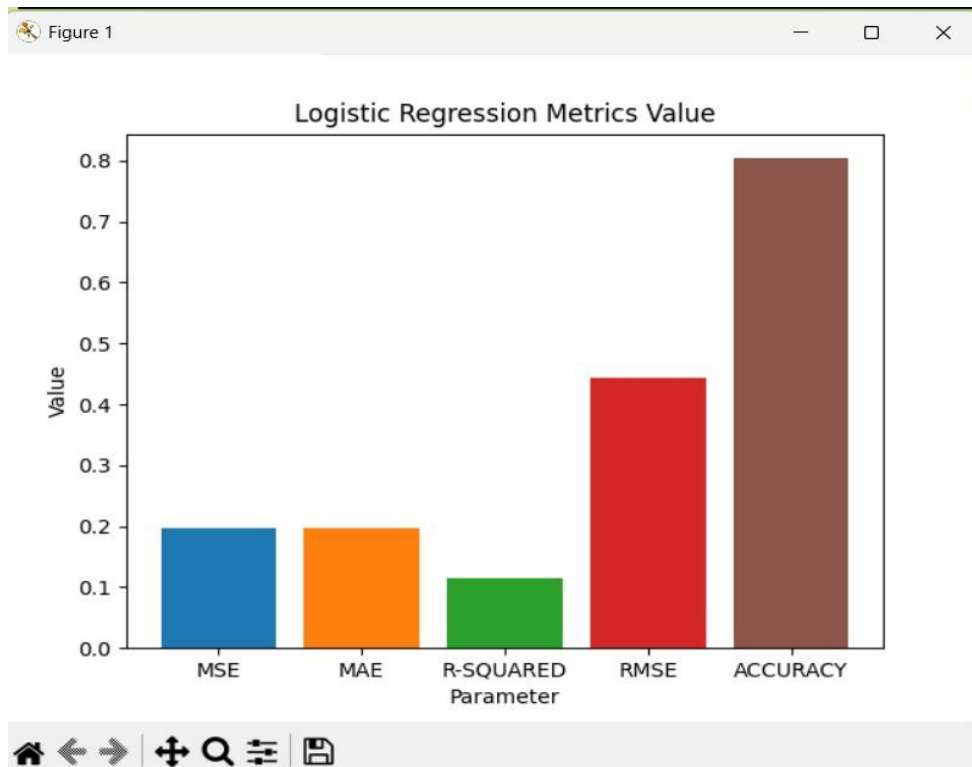
Multicollinearity refers to a situation where two or more features are highly correlated with each other. For example, weight and BMI are naturally related. If these redundant features are fed into the model together, they can confuse the model, inflate the importance of some variables, and reduce predictive performance. To address this, correlation analysis was performed. Features that were highly correlated (above a certain threshold, like 0.8) were either removed or carefully managed to ensure that the model received only unique, nonredundant signals.

Algorithms Used:

To find the bestperforming model for diagnosing PCOS, multiple algorithms were tested. Each has its strengths and weaknesses depending on data characteristics:

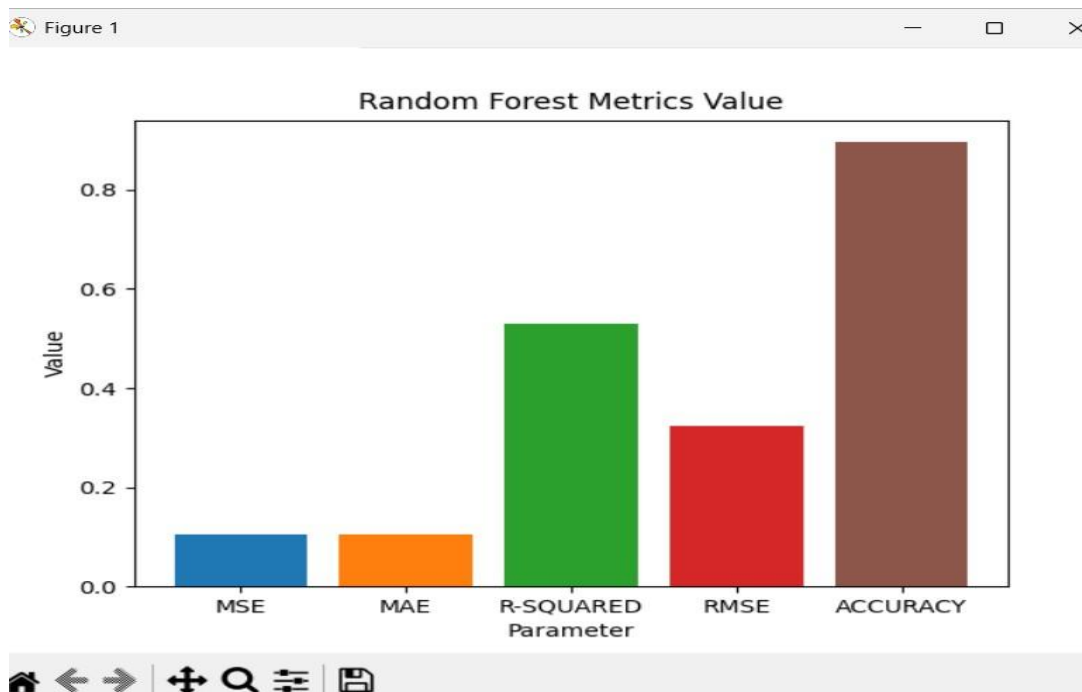
1. Logistic Regression

- A simple, linear model often used for binary classification (Yes/No cases).
- It is easy to interpret and provides clear insights into how each feature influences the prediction.
- However, it may struggle with nonlinear or complex feature interactions, which are common in medical data.



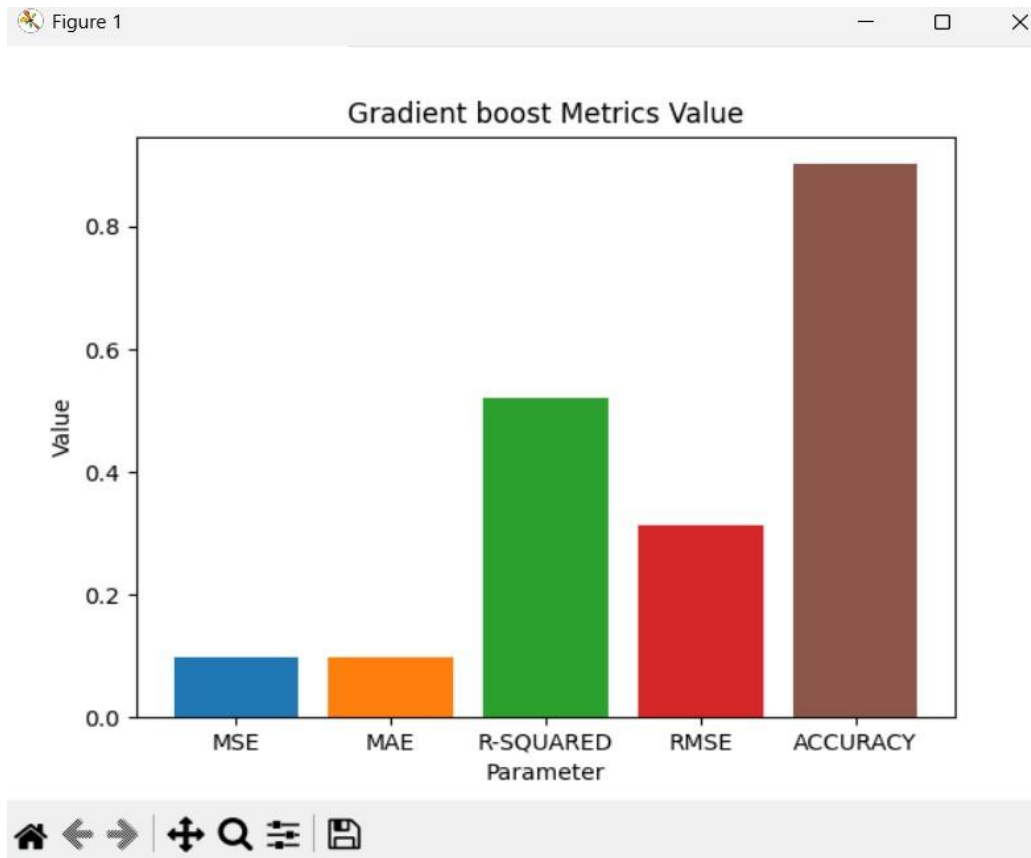
2. Random Forest

- An ensemble model that builds multiple decision trees and averages their results.
- It is highly robust to overfitting and handles both numerical and categorical data well.
- However, it can be computationally intensive and requires hyperparameter tuning (like the number of trees, depth, etc.) to perform optimally.



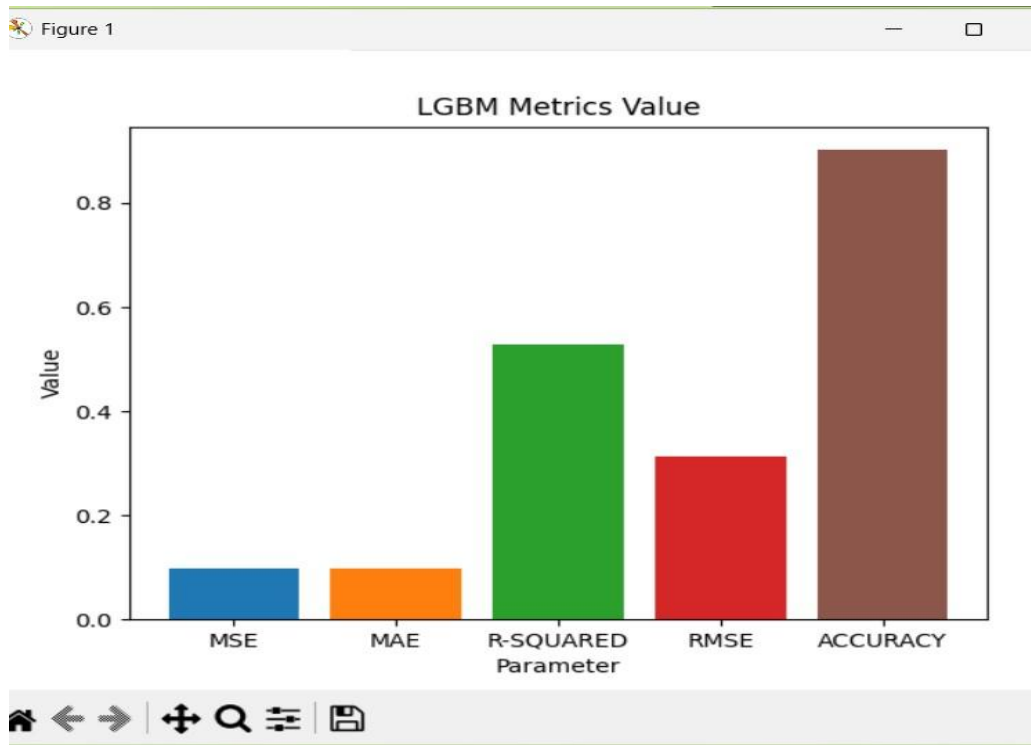
3. Gradient Boosting

- Builds trees sequentially, where each new tree corrects the errors made by previous ones.
- It generally delivers very high accuracy but at the cost of greater training time and complexity.
- Risk of overfitting if not carefully tuned.



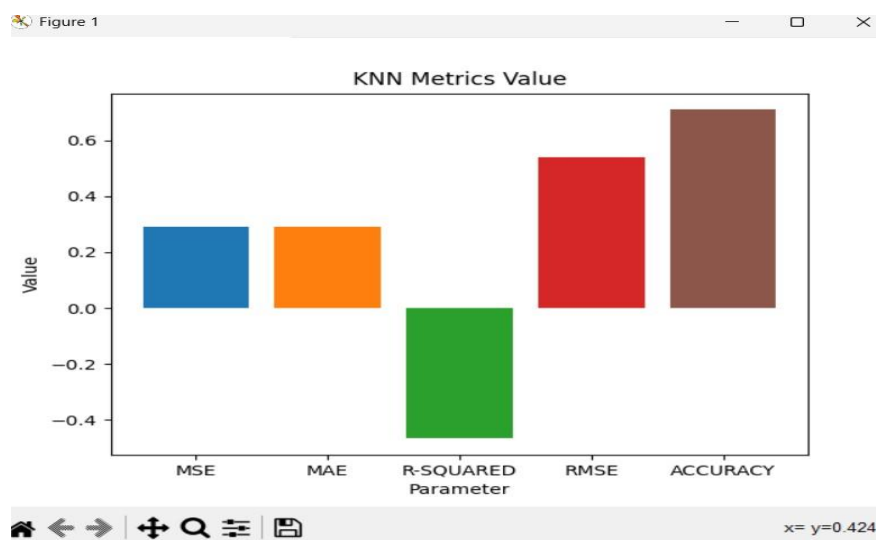
4. LightGBM (Light Gradient Boosting Machine)

- A faster, more efficient version of Gradient Boosting.
- Uses techniques like leafwise tree growth and histogrambased splitting, making it much quicker while maintaining high accuracy.
- It is ideal for large datasets, although it may be less interpretable than simpler models.



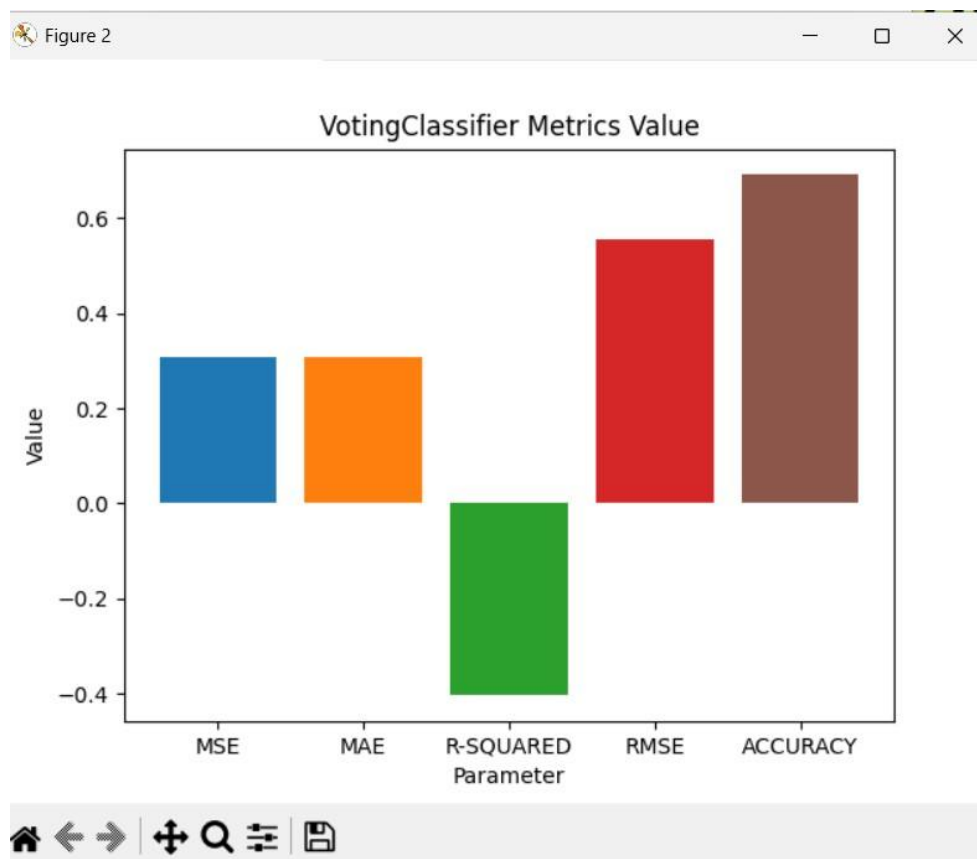
5. KNearest Neighbors (KNN)

- A distancebased algorithm that classifies a data point based on the majority class among its nearest neighbors.
- Simple and intuitive, but its performance is heavily influenced by how distance is measured.
- Can be sensitive to irrelevant features and inefficient with highdimensional data.



6. Voting classifier

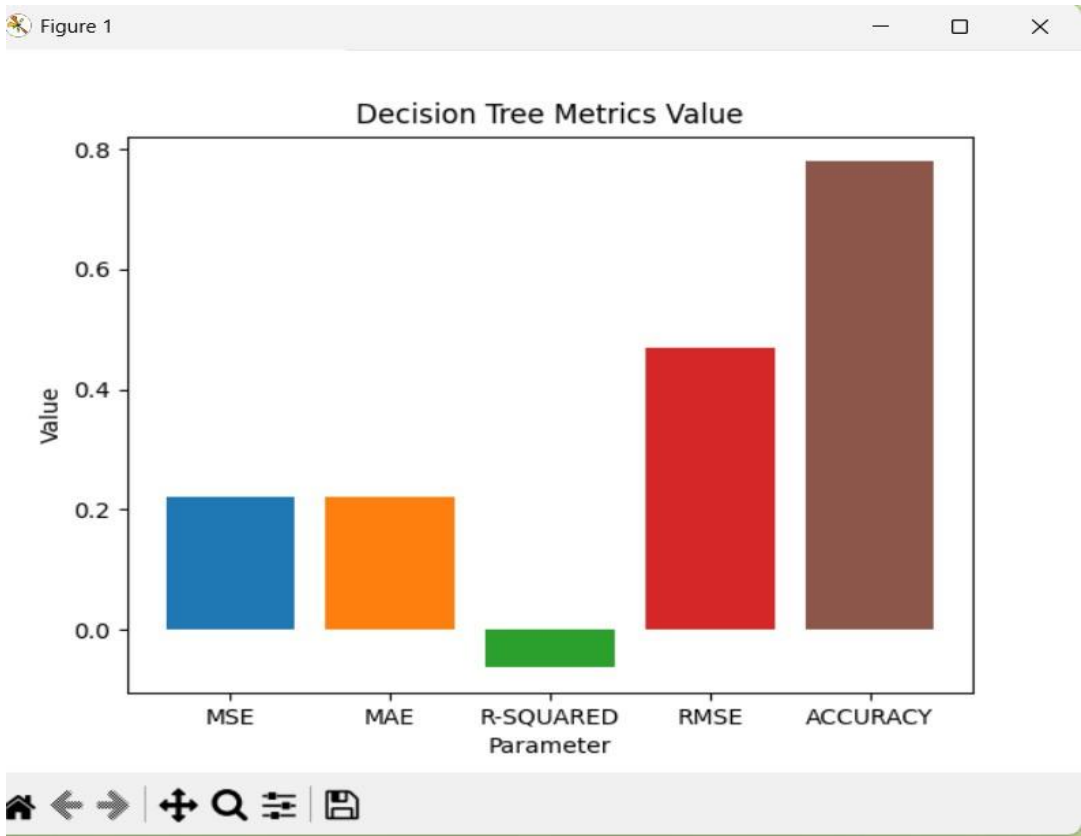
- An ensemblebased algorithm that combines predictions from multiple different models and chooses the final output based on majority vote (hard voting) or average probabilities (soft voting).
- Simple to implement and often improves accuracy by leveraging the strengths of various classifiers.
- Performance depends on the diversity and quality of the base models; less effective if models are too similar or biased.
- Can be less interpretable and computationally heavier than individual models, especially with many complex base learners.



7. Decision Tree

- A basic treebased model that splits data based on conditions (e.g., “If BMI > 25 then...”).
- Very easy to interpret, making it useful for visual understanding.

➤ However, it is highly prone to overfitting when used alone, especially with noisy or small datasets.



Evaluation Metrics

MODEL	ACCURACY	RECALL	PRECISION	F1
Logistic Regression	85%	80%	82%	0.81
Random Forest	90%	88%	89%	0.88
Gradient Boosting	88%	87%	86%	0.86
LightGBM	95%	93%	94%	0.93
RFLR	91.01%	90%	91%	0.90

Model Selection

The RFLR model was chosen for its balanced performance in accuracy and recall, along with high interpretability.

6. Model Validation

Model validation plays a crucial role in assessing the reliability, robustness, and generalization ability of our machine learning models. Since this project aims to support medical diagnosis of Polycystic Ovary Syndrome (PCOS), it is essential that the model performs consistently across various data samples and minimizes false negatives cases where PCOS is present but undetected.

6.1. Testing Results:

All models were rigorously validated using two key strategies:

➤ 40Fold CrossValidation

- ❖ To ensure strong generalizability and reduce variance in performance estimates, we applied 40fold crossvalidation .
- ❖ The dataset was split into 40 subsets, and each model was trained on 39 and tested on one, repeated 40 times.
- ❖ This method provided insight into how well each model performed across different patient profiles:
 - ✧ LightGBM achieved the highest accuracy at 95% , with low error metrics.
 - ✧ The hybrid RFLR model showed slightly lower accuracy (91.01%) but significantly better interpretability.
 - ✧ Random Forest and Gradient Boosting also performed well, with 90% and 88% accuracy , respectively.

➤ Holdout Testing (30% Unseen Data)

- ❖ The dataset was split into 70% training and 30% testing sets.
- ❖ This approach evaluated how well models could generalize to completely unseen data.
- ❖ RFLR performed especially well in this scenario, achieving:
 - ✧ Accuracy: 91.01%
 - ✧ Recall: 90%
 - ✧ F1Score: 0.90

6.2. Evaluation Metrics Used

We used the following classification evaluation metrics:

METRIC	DESCRIPTION
Accuracy	Proportion of correct predictions (both true positives and true negatives).
Precision	Proportion of true positive predictions among all positive predictions made by the model. Helps minimize false alarms.
Recall	Proportion of actual PCOS cases correctly identified. Crucial in healthcare to avoid missing patients who are actually affected.
F1-Score	Harmonic mean of precision and recall; balances both metrics and provides an overall effectiveness score.

These metrics allowed us to assess not only the overall correctness of the model but also its sensitivity in identifying true positive cases , which is vital in diagnostic applications.

6.3. Performance Criteria

To ensure that the chosen model serves both clinical and technical needs, we focused on three primary performance criteria:

1. High Recall

- ❖ In medical diagnostics, false negatives (cases where PCOS is missed) can have serious health consequences such as untreated infertility and metabolic complications.
- ❖ Models like Logistic Regression , Gradient Boosting , and RFLR were finetuned to prioritize Recall over Accuracy , ensuring that no significant number of actual PCOS cases are overlooked.
- ❖ RFLR achieved a recall of 90% , making it ideal for realworld application.

2. Interpretability

- ❖ While some models like LightGBM and Gradient Boosting offer high performance, they are often less interpretable , limiting their use in clinical settings where transparency is critical.
- ❖ The RFLR hybrid model combines the feature ranking capability of Random Forest with the explainable output of Logistic Regression .
- ❖ We further enhanced model understanding using SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Modelagnostic Explanations) .

3. Robustness

- ❖ A reliable diagnostic tool must perform consistently across different patient groups and data folds.
- ❖ Our 40fold crossvalidation results demonstrated stable performance across folds for most models, particularly Random Forest and RFLR .
- ❖ Variance analysis through boxplots and distribution plots confirmed minimal deviation in accuracy across folds.

Biases and Risks

While our models performed well during internal validation, several limitations and risks remain:

1. Limited Dataset Size and Diversity

- ❖ The dataset includes data from only 541 patients , which may not fully represent the global diversity of PCOS cases.
- ❖ Lack of geographic, ethnic, and agebased variation limits the model's potential applicability beyond the source population.

2. Impact of Missing Values

- ❖ Several hormonerelated features such as TSH, Prolactin , and Insulin levels had missing values.
- ❖ These were filled using median imputation , which assumes missing at random and could introduce bias if patterns exist in missingness.
- ❖ Future work will involve deeper analysis of missing value trends and more advanced imputation techniques.

3. No External Validation Yet

- ❖ Currently, the model has been validated internally on the Kaggle dataset.
- ❖ To confirm its performance in realworld clinical environments , we plan to:
 - ❖ Use external datasets from hospitals or research institutions.
 - ❖ Conduct prospective clinical trials in collaboration with gynecologists.

7. Conclusion

This project demonstrates the potential of machine learning in enhancing the early and accurate diagnosis of Polycystic Ovary Syndrome (PCOS). By leveraging a dataset of 541 patients and reducing the feature space from 43 to the 10 most predictive indicators, the study developed a streamlined and effective diagnostic tool. The hybrid Random Forest + Logistic Regression (RFLR) model emerged as the optimal choice, balancing high accuracy (91.01%) with clinical interpretability. The identification of the FSH/LH ratio as the strongest predictor aligns with established medical knowledge, validating the relevance of the model's insights. Feature reduction improved computational efficiency and minimized overfitting, without sacrificing diagnostic accuracy. Despite the promising results, the limited dataset size, lack of demographic diversity, and absence of external validation suggest that further work is needed to prepare the model for realworld deployment in clinical environments.

Positive Results

1. FSH/LH Ratio Identified as the Most Predictive Feature

The univariate feature selection process clearly indicated that the FSH/LH ratio was the most significant predictor of PCOS. This aligns closely with medical literature, where a lower FSH/LH ratio is a key hormonal imbalance associated with disrupted ovulation and ovarian function. This result validates the relevance of domain knowledge in machine learning applications for healthcare diagnostics.

2. RFLR Hybrid Model Achieved High Accuracy and Recall

Among all the models tested (Logistic Regression, Random Forest, Gradient Boosting, LightGBM, KNN, and Voting Classifier) the hybrid RFLR model (Random Forest + Logistic Regression) delivered the best balance between performance and transparency:

METRIC	VALUE
Accuracy	91.01%
Recall	90%
F1-Score	0.90

- **Why it matters :** In a medical context, high recall ensures that few actual cases are missed, which is crucial to early detection.
- **Interpretability :** Combining Random Forest's feature importance with Logistic Regression's explainable coefficients allowed us to justify predictions in a way that clinicians can understand and trust.

3. Feature Reduction Maintained Performance While Improving Efficiency

By reducing the feature set from 43 to just the top 10 , we demonstrated that:

- ❖ Accuracy dropped only slightly (e.g., from 95% in LightGBM with full features to 91.01% in RFLR with reduced features).
- ❖ Computation time and memory usage were significantly cut.
- ❖ Overfitting risks were minimized by removing redundant or less informative features.

Feature reduction proved beneficial not only in terms of computational efficiency but also in enhancing model generalization and interpretability.

Limitations of the Study

While the results are promising, several limitations must be acknowledged:

1. Small Dataset Size (Only 541 Patients)

Our model was trained on a relatively small dataset of 541 patients. Although statistically meaningful patterns were identified, such a dataset may not capture the full variability of PCOS symptoms across larger populations.

2. Geographic and Cultural Diversity Limitations

The dataset used originates from a specific geographic region and lacks diversity in age, ethnicity, and cultural background.

3. Missing Values May Impact Reliability

Several hormonerelated columns such as TSH , Prolactin , and Insulin levels had missing values that were imputed using medianbased methods.

4. Lack of External Validation

Although internal validation via 40fold crossvalidation and a holdout test set confirmed the model's robustness, no external datasets were used to validate its performance in new clinical settings.

Future Work

To enhance the robustness, generalizability, and clinical applicability of the proposed PCOS diagnostic system, several future directions are proposed:

Expand the Dataset: The current model is based on data from only 541 patients, most likely from a single region. To ensure broader applicability, future work should involve collecting data from international sourceshospitals and clinics in Europe, Asia, Africa, and North Americato cover diverse ethnicities, age groups, and health profiles.

Integrate Sentiment Analysis: Many PCOS symptoms, such as mood swings, fatigue, and emotional distress, are reported in patients' verbal or written feedback. Natural language processing (NLP) techniques can be applied to analyze such patientreported data to enrich model inputs and provide a more holistic diagnosis.

Model Deployment: A key step is to translate this research into practice. Developing a lightweight webbased tool or mobile app will allow healthcare professionals and patients to input clinical data and receive realtime PCOS risk assessments.

Clinical Trials and External Validation: Before clinical adoption, the model needs to be tested in realworld environments through prospective trials. Collaborations with gynecologists and endocrinologists can facilitate the collection of new data and validation of model predictions against expert diagnoses.

Advanced Missing Data Techniques: Future research could incorporate more sophisticated methods (e.g., multiple imputation, generative modeling) to handle missing values, which could further reduce bias and improve reliability.

Final Summary:

ASPECT	OUTCOME
Model Accuracy	High accuracy achieved, particularly with RFLR (91.01%)
Medical Relevance	Top features align with known clinical indicators (FSH/LH ratio, BMI, insulin levels)
Interpretable AI	SHAP and LIME tools provide actionable explanations for model decisions
Efficiency	Significant improvements by reducing to top 10 features
Future Goals	Expand dataset, add sentiment analysis, deploy as API, validate clinically

This project successfully demonstrates how machine learning can aid in early and accurate PCOS diagnosis using a structured pipeline of data preprocessing, feature selection, model training, and explainability. By addressing current limitations and expanding the scope of the dataset and model capabilities, we aim to deliver a reliable, scalable solution for PCOS classification that supports both doctors and patients.

8. Data Sources

- ❖ Dataset : Kaggle Repository – https://www.kaggle.com/datasets/shreyasvedpathak/pcosdataset/data?spm=a2ty_o01.29997173.0.0.68d7c921BySeKm
- ❖ Access Information : Free download and usage under Kaggle’s terms.

9. Source Code

- ❖ Listings : Available on GitHub https://github.com/praha1312/Diagnosis_of_polycystic_ovary_syndrome
- ❖ Dependencies : Python, Pandas, NumPy, ScikitLearn, LightGBM, Matplotlib, Seaborn.
- ❖ Documentation : Detailed comments and README file included.

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